Exercise: Configure aggregations using DAX

Introduction

In your experience with Power BI, you've learned to turn data into insights. In this exercise, you'll use **DAX** to calculate annual and quarterly profits and analyze data sets to create a clear financial story.

By completing this exercise, you'll demonstrate your ability to:

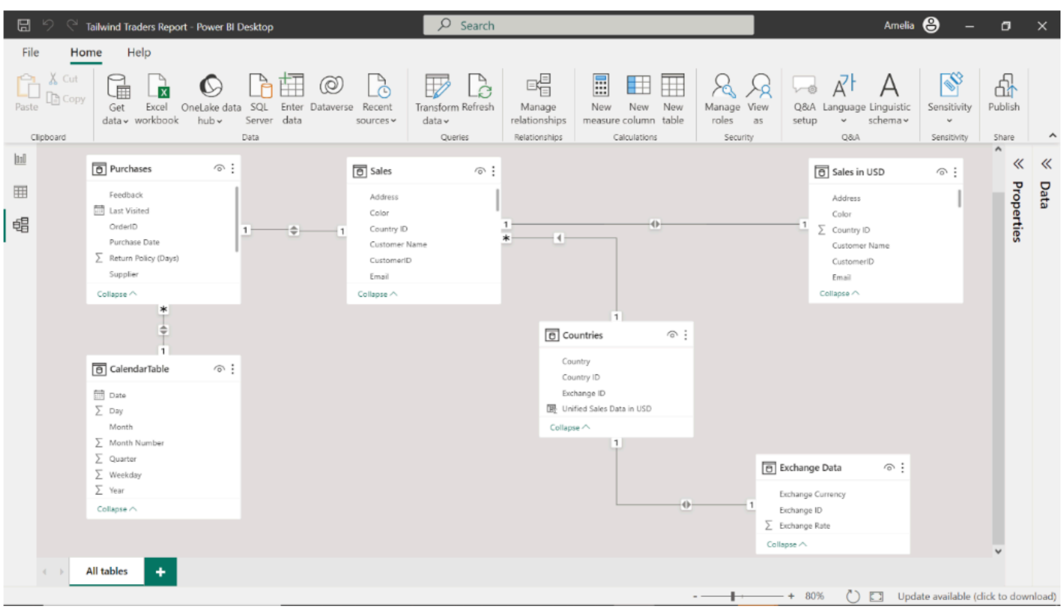
- **Create time-based summaries for displaying quarterly, annual, and year-to-date profit data.**
- **Determine median sales volume to assess Tailwind Traders' performance stability.**
- **Utilize the Performance Analyzer tool to enhance report generation and ensure fast loading times.**

Scenario

Tailwind Traders needs to generate insights into its financial performance to inform its strategic decisions for the upcoming business year. The insights it requires include time-based summaries that display quarterly and annual profits and year-to-date breakdowns. The company also requires its median sales volume to assess its financial performance. Let's help Tailwind Traders create this report.

Instructions

Locate and open the **Tailwind Traders Report.pbix** Power BI file you created in the previous exercise. If you followed the steps from the previous exercise correctly, then your data model should resemble the one displayed in the screenshot below. Follow the prompts to configure aggregations for this data model using DAX.



Step 1: Calculate Yearly Profit margin

1. Create a new measure for the **Sales in USD** table.
2. In the formula bar, create a new measure using a **DAX** function that represents the yearly profit margin. This margin should be calculated by dividing total profit by net revenue within the **Sales in USD** table.
3. In the **Fields** pane, select the new measure and change its format to **Percentage in the Properties** pane.

Tip: Consider using **DIVIDE** and **SUM** functions to create this ratio of profit to revenue.

Step 2: Calculate Quarterly Profit

1. Right-click on the **Sales in USD** table in the **Fields** pane and choose **New Measure**.
2. In the formula bar, create a new measure for quarterly profit using the **Profit in USD** column. You'll need a function that aggregates data until the end of the current quarter, referencing both the profit values and a calendar table.
3. Format the new measure as a **Percentage**.

Tip: Consider using the **DATESQTD** function with your profit values to aggregate data by quarter.

Step 3: Calculate Year-to-Date Profit

1. Right-click on the **Sales in USD** table in the **Fields** pane and select **New Measure**.
2. In the formula bar, create a new measure for the year-to-date profit using the **Profit in USD** column. You'll need a function that aggregates profit data from the start of the year up to the current date.
3. Format the new measure as a **Percentage**.

Tip: Consider using **TOTALYTD** with your profit values and calendar table to calculate running totals from the start of each year.

Step 4: Calculate Median Sales

1. Right-click on the **Sales in USD** table in the **Fields** pane and choose **New Measure**.
2. In the formula bar, create a new measure to represent the median sales based on the **Gross Revenue USD** column. Consider which statistical functions in **DAX** can help you find the middle value of your sales data.

Tip: Consider using the **MEDIAN** function to find the central value in your **Gross Revenue USD** data. The median separates the higher and lower half of a data sample.

Step 5: Access the Performance Analyzer

1. Find and select the **Performance Analyzer** option within the **View** tab.
2. Create an empty **Card** visual and drag the **Yearly Profit Margin** field to the **Fields** well. Repeat this process for the **Median Sales**, **Quarterly Profit**, and **YTD Profit**.
3. Begin recording the performance of the card visuals using the Performance Analyzer's recording feature.
4. Refresh your reports to test their performance.
5. Select the plus (+) symbol next to each **Card** to open up the details.
6. Observe the list of all visual items in your report and their respective load times. Ensure the **DAX** query time of visual items is < 200ms and note any slow-loading visuals.
7. Select **Stop** and remove all **Card** visuals, resulting in a blank Canvas.
8. Save your report.

Conclusion

You've helped influence Tailwind Traders' strategic decisions by calculating profits and assessing sales consistency. In the data world, accuracy and speed are key. You've also demonstrated the ability to configure calculations using DAX.